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Predicting Reward-Based Crowdfunding Success with Multimodal Data: A Theory-Guided Framework

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Abstract: There is a growing need to investigate the impact of multimodal data, which are becoming increasingly prevalent on crowdfunding platforms, on prediction of fundraising outcomes. However, a prediction framework drawing upon rational theoretical foundations to leverage multimodal data in crowdfunding is still lacking. Guided by relevant theories, we explore the ideational, interpersonal,

and textual metafunctions of multimodal data geared toward fundraising success prediction. Our empirical evaluation demonstrates superior predictive utilities of various metafunction-based multimodal features over purely data-driven ones. Our results also reveal that the multiple data modalities interact complementarily and synergistically to improve the prediction performance. Specifically, combining metafunctions improved prediction performance by 2–15% over a single metafunction, while multimodality outperformed single data modality by 7–18% within each metafunction.

Keywords: crowdfunding success prediction; multimodal data; metafunction; large language models; deep learning.

1 Introduction

Multimodal data, when various forms of data, such as linguistic, visual, and acoustic data, are collected and used in the prediction process, can enhance a prediction model's performance (Norris and Maier 2014). In the information systems (IS) community, leveraging multimodal data in predictive analytics has emerged as one of the most popular research topics, as it provides a broader perspective on characterizing the target business scenario. A large number of applications in IS contexts, such as social media, e-commerce, and healthcare informatics, involve the integration of multimodal data to provide a more accurate, comprehensive, and contextually rich representation of the prediction task. Especially as multimodal intelligence and its applications, such as multimodal large language models and multimodal agents, become increasingly prevalent, leveraging multimodal data for business prediction tasks shows prominent research prospects.

Crowdfunding is an alternative finance (Ziegler et al. 2019) and extension of crowdsourcing (Schweinbacher and Larralde 2012) that allows fundraisers to secure small amounts of funding from

large audiences, as opposed to raising substantial sums from a small group of investors (Belleflamme et al. 2014). This approach is typically supported by four primary models: lending (P2P lending)-, equity-, reward-, and donation-based crowdfunding (Mollick 2014; Belleflamme et al. 2014; Shneor et al. 2020). Lending-based crowdfunding involves individual or institutional backers providing loans to borrowers, expecting repayment of the principal and interest within a specified period. Equity-based crowdfunding allows backers to purchase ownership stakes in an organization. Reward-based crowdfunding offers backers non-monetary rewards, typically in the form of pre-purchased products or services, in exchange for their financial support. Donation-based crowdfunding is based on philanthropic or civic motivations, where backers do not expect material returns.

Among these models, reward-based crowdfunding is particularly reliant on multimodal data to assist backers in making informed decisions. In lending-based crowdfunding, successful loans are mainly associated with the characteristics of borrowers and loans, such as credit scores (Kuwabara et al. 2017; Pope and Sydnor 2011), debt-to-income ratio (Larrimore et al. 2011; Pope and Sydnor 2011), and interest rates of loans (Feng et al. 2015; Larrimore et al. 2011). Equity-based crowdfunding typically involves a longer time horizon for investors to realize returns and carries a higher risk of capital loss, especially in early-stage ventures. Therefore, potential investors assess the quality of the venture and the entrepreneur's human capital (education and entrepreneurial experience) to make their decisions. As for donation-based crowdfunding, it is driven by donors' altruistic motivations (Choy and Schlagwein 2015), with a tendency to favor campaigns that align with their location or ideology. In summary, funders' actions in lending-, equity-, and donation-based crowdfunding models are primarily influenced by the characteristics of fundraisers (such as expertise, experience, and fundraising history) or static information about the campaigns (such as funding goals, interest rates, and timelines). In contrast, in reward-based crowdfunding, rewards are products or services that are

most likely in the conceptual or production stage. Backers place significant importance on the detailed descriptions of the products or services, where multimodal data play a crucial role. Therefore, we have selected reward-based crowdfunding as the focal context for evaluating our multimodal analytical framework.

Despite the rapid expansion of the reward-based crowdfunding market, successfully acquiring the target amount of a project remains a significant challenge. According to a survey conducted on a popular reward-based crowdfunding platform, of all the posted projects, only 40.97% ended up fully funded.¹ Platforms typically adopt an *all-or-nothing* policy, where entrepreneurs receive the pledged amount only if they reach their funding goals within the fundraising period. If projects fail, neither entrepreneurs nor backers receive any benefit at all. *Therefore, predicting the likelihood of success throughout the fundraising period is of great value for both entrepreneurs and backers.*

While quantitative information (e.g., the presence and number of descriptive texts, images, or videos) has been investigated extensively by previous crowdfunding studies (Carradini and Fleischmann 2023; Mollick 2014; Yang et al. 2020; Zhou et al. 2018), the impact of unstructured multimodal data (e.g., linguistic, visual, and acoustic features) on the fundraising outcome has been largely overlooked. Yet the utilities of various modalities of unstructured data have been revealed in other contexts, such as hyperspectral image and light detection and ranging (Yang et al. 2024) in image recognition and classification, grid and graph modals in mobile traffic prediction (Jiang et al. 2024), semantic devices (Musté et al. 2015) and visual features (Riegelsberger et al. 2002; Chan and park 2015; Garrett 2002) in Internet marketing, and colors (Chan and Park 2015), hedonic features (Riegelsberger 2002; Riegelsberger 2003; Cyr et al. 2009), and acoustic features (Short et al. 1976; Zhu et al. 2010) in e-commerce. Reward-based crowdfunding shares many similar characteristics with e-commerce, where

¹ <https://www.kickstarter.com/help/stats>.

a retailer or an entrepreneur posts descriptive texts, images, or videos to introduce their products or services. Since customers cannot receive an actual product or service before their payments, and they are unlikely to receive consultations from financial institutions for evaluating, monitoring, and managing risks (Huang and Knight 2017; Kim and Viswanathan 2019), multimodal information offers an effective channel to visualize the product or service. The combination of multiple modalities can convey a panoramic view of products or services, including colors, movements, feelings, and emotions, to potential backers more effectively than plain texts. This expansion of product or service description can help potential backers make better informed decisions. Therefore, the incorporation of multimodal information has great potential not only for enhancing the trustworthiness of fundraising projects but also for offering panoramic predictive views for reward-based crowdfunding success.

However, effectively leveraging multimodal data for crowdfunding success prediction is challenging because directly combining the linguistic, visual, and acoustic content of multiple modalities incurs the curse of high feature dimensionality, where noise and redundant features tend to weaken the performance of multimodal prediction. The exploration of how multiple data modalities interact complementarily and synergistically to play a pivotal role in predicting fundraising success is still lacking. Further, given various data modalities, it is also difficult to determine which features and feature modalities should be constructed and whether they are sufficient to reveal the prediction target. An effective way to leverage multimodal data for crowdfunding success prediction is to develop a multimodal prediction framework drawing upon rational theoretical foundations (Feng 2011).

Hence, in this study, we strive to probe the following core research questions:

1. What theoretical frameworks from the IS area can be applied to analyze various semiotics in multimodal data?

2. How can a multimodal prediction framework be constructed drawing upon rational theoretical foundations?
3. How can multimodal data be leveraged appropriately and rationally to enhance reward-based crowdfunding success prediction?
4. What is the effectiveness of different data modalities in prediction models for reward-based crowdfunding success?

To this end, we investigate the complementary and synergistic interactions among linguistic, visual, and acoustic clues on reward-based crowdfunding platforms, aiming to identify the metafunctions of multimodality. We gauge the metafunctions and expand the range of data modalities to improve fundraising success prediction performance. Our empirical evaluation results, based on data from Kickstarter.com, show that various data modalities yield distinct predictive values when conveying different metafunctions. The dynamic pattern implies that it is crucial to consider the metafunctions in leveraging multimodal data for fundraising success prediction.

Our study makes multiple contributions to the IS knowledge base.

1. We contribute to the literature on multimodality in IS. Our study pioneers the suggestion of systemic functional linguistics (SFL) and its extension to visual design as a robust theoretical foundation for studying IS applications where multimodal data interact complementarily to achieve a task. We propose a novel framework by identifying representation features of metafunctions from text, image, and video modalities, grounded in this theoretical foundation.
2. Our study enriches the literature on crowdfunding success prediction. We develop predictive models that assess the predictive capability of each metafunction separately. Our empirical results highlight the effectiveness of metafunctions in revealing the interactions among

multiple data modalities within a single page, thereby providing coherent intersemiotic² information and showing superior predictive utilities over the data-driven multimodal model without considering metafunctions.

3. Our study demonstrates the importance of increasing data modalities in improving prediction performance. We also uncover patterns associated with different data modalities in predicting fundraising success, shedding light on the relationships among modalities and their impact on predictive outcomes.

2 Literature Review

2.1 Determinants of Fundraising Success

Crowdfunding has emerged as a prominent avenue for entrepreneurs to obtain capital from the public via online platforms. Given that all the stakeholders (entrepreneurs, backers, and platforms) hope to achieve success, a substantial body of research has investigated determinants that influence fundraising performance in terms of different measures, such as the number of backers, funds raised, funding success, and success ratio (Deng et al. 2022). The determinants are typically categorized into four types: project-, creator-, backer-, and platform-related factors (Koch and Siering 2019; Kaartemo 2017; Deng et al. 2022; Zhou et al. 2018).

Project-related determinants involve the intrinsic characteristics of funding campaigns (Cumming et al. 2015). The commonly investigated determinants include funding goal, funding duration, reward levels, project category, comments, and numbers of images, videos, FAQs, and updates. Generally, funding goal and duration are negatively correlated with crowdfunding success (Bi et al. 2017; Josefy et al. 2017; Zheng et al. 2014; Mollick 2014; Davis et al. 2017). The presence and quality of images and videos, FAQs, and updates tend to influence success positively (Bi et al. 2017; Hobbs et al. 2016;

² Intersemiotic refers to two or more different codes, such as linguistic sign systems and image ones.

Liang et al. 2020; Mollick 2014; Yang et al. 2020). Project-oriented comments achieve better crowdfunding performance than investor-oriented comments (Wang et al. 2024). Sold-out reward options help in increasing funding revenues (Wessel et al. 2019). The category is usually viewed as a control variable. Greenberg and Mollick (2017) found that publishing and fashion-related projects are more likely to succeed.

Creator-related factors are associated with the demographics and backgrounds of the individuals or organizations initiating projects. These include gender (Allison et al. 2017; Duan et al. 2020; Von and Lutz 2021), cultural background (Josefy 2017), race (Anglin et al. 2018; Davis et al. 2017), hedonic value (Zhao and Vinig 2017), entrepreneurial experience (Cornelius and Gokpinar 2020), social media connections (Skirnevskiy et al. 2017; Courtney et al. 2017; Buttice et al. 2017; Mazzocchi and Lucarelli, 2022), and self-funding initiatives (Zhao et al. 2022b). In most cases, these factors serve as control variables.

Backer-related factors focus on the motivations and interactions of the individuals supporting projects. For example, motivation is an important driver of behavior in reward-based crowdfunding (Steigenberger 2017). Backers' co-creation and psychological ownership motivate them to contribute to a project (Zheng et al. 2018). Prosocial motivation arises in driving backers' pledging and sharing decisions as the pledged amount is approaching the funding goal (Li and Wang 2019). Altruistic motives have been shown to enhance the likelihood of a campaign's success (Giudici et al. 2018), and positive interactions between entrepreneurs and backers correlate with higher success rates (Shnor et al 2023; Zhao et al. 2024).

Platform-related factors examine aspects including the number of projects being funded on the platform (Chan et al. 2021), the model of the platform (Cordova et al. 2015), and the number of years since the platform's establishment (Shneor et al. 2020). For example, "All-or-Nothing" and "Keep-

It-All” are two typical fundraising models, which also impact the likelihood for a project to meet its fundraising target. Specifically, Keep-It-All campaigns typically exhibit lower success rates in achieving their financial goals (Cumming et al. 2015; Josefy 2017).

While this comprehensive examination of the literature highlights the prominent hard factors that can influence crowdfunding success, soft information extracted from multimodal data also plays a vital role in reward-based fundraising success.

2.2 Systemic Functional Linguistics and Metafunction

Systemic functional linguistics (SFL) was first developed by the linguist Michael Halliday in the early 1960s as a tool for teaching Mandarin (Halliday 1985) and later extended to the English language (Halliday 1994). It serves as the central theoretical framework for systemic functional approaches to multimodality, analyzing the function and meaning of semiotic resources (O’Halloran 2008a). Semiotic resources are theorized as realizing three different meaning functions, known as *metafunctions*. This metafunctional system is used to interpret how semiotic resources simultaneously construct experiences and logic (*ideational* meaning), enact social relations (*interpersonal* meaning), and organize a structured text (*textual* meaning).

Drawing on the insights of Halliday, researchers have gradually extended SFL to non-verbal semiotic resources and media, such as visual design (Kress and Van Leeuwen 1996), advertisements (O’Halloran, 2008b), websites (Djonov 2005; Bateman 2008), and music (Van Leeuwen 1999). Kress and Van Leeuwen (1996) argued that the same metafunctions used in linguistic analysis can be applied to visual resources. The three metafunctions—ideational meaning, interpersonal meaning, and compositional (textual) meaning—can be identified in visual imagery. The term “compositional” has been used as a substitute for Halliday’s term “textual” to better convey the idea of two modes interacting with each other within one page to provide a coherent intersemiotic message (Royce 2007).

SFL and its metafunctions provide a comprehensive functional representation of language and visual design. As a result, they have become fundamental theories in the areas of linguistics, communication, education, and social sciences. However, they are rarely used as root theories in the IS area. Although some literature has studied valuable feature capture referring to SFL, such as computer-mediated communication (Abbasi and Chen 2008), social media (Dong et al. 2018), and tacit knowledge elicitation (Zappavigna and Patrick 2010), prior studies extract text features only, overlooking their power in dealing with multimodal data. Therefore, we aim to explore whether identifying and extracting metafunctions from multimodality can help to solve IS problems. Given that reward-based crowdfunding websites provide multiple semiotic modes, we analyze multimodal data based on the metafunctions theory to predict fundraising success.

2.3 Multimodality in Crowdfunding

Besides structured project-related characteristics, researchers have increasingly focused on unstructured and qualitative information conveyed through multimodal data—text, images, and video—that enrich project descriptions and facilitate informed decision-making by potential backers. A substantial body of research highlights the influence of various data modalities on fundraising outcomes. The most commonly investigated modality is text. On one hand, overall textual characteristics, such as length (Adamska-Mieruszevska et al. 2021, Zhou et al. 2018), readability (Adamska-Mieruszevska et al. 2021, Zhou et al. 2018), and lexical sophistication (Mahmood and Yeganegi 2023), have been shown to influence fundraising success. On the other, specific factors, elements, and features conveyed in texts, such as informational or emotional appeals (Bao et al. 2022; Xiang et al. 2019), sentiment and topic (Jiang et al. 2020; Wang et al. 2017; Yuan et al. 2016), title and blurb (Wang et al. 2020) words showing entrepreneurs' personality (Anglin and Pidduck 2022), semantics (Bao et al. 2022), social media connection (Li and Wang 2019), and narratives (Naimi et

al. 2023; Tajvarpour and Pujar. 2022.; Wang et al. 2020; Wang et al. 2021; Zhu, 2022), also influence campaign outcomes.

Moreover, visual elements in crowdfunding also yield considerable impact. The disclosure of entrepreneurs' faces reduces the amount raised, whereas the appearance of smiling faces increases the fundraising amount (Kim and Park 2017) and mitigates overhead aversion (Yoo et al. 2022). Beyond mere appearances, visual attributes, such as color (Chan and Park 2015; Barnes 2024), social cues (Barnes 2024), emotions (Hou et al. 2019, Raab et al. 2020), and contexts (Ma and Palacios 2021), are significant predictors of success. Additionally, human appearance in images (Anderson and Saxton 2016), logo complexity (Mahmood et al. 2019), product complexity, owner appearance, and exemplification in videos (Dey et al. 2017; Kaminski et al. 2017; Korzynski et al. 2021) can help to predict fundraising outcomes.

Furthermore, multimodal data foster trust between fundraisers and potential backers, especially performance/cognition-based trust (Liang et al. 2019), enhancing potential backers' motivations in funding projects. For example, high-quality project information, persuasive narratives, and indications of the fundraiser's capability and reputation have significant impacts on trust (Liang et al. 2019). Factors such as supporting calls for funding action, social media shares, word sentiment, and topic trends provide valuable insights into social media engagement and content quality, which are two critical trust-enhancing elements of crowdfunding (Baah-Pepurah and Shneor 2022). Face-to-face activities, photos, blogs, and updates also play essential roles in establishing benevolence-based trust, which further increases fundraising success (Lui et al. 2023). Additionally, an entrepreneur's facial creditworthiness and entrepreneur-backer interactions promote fundraising performance (Zheng et al. 2016; Duan et al. 2020).

Although existing studies have significantly advanced our understanding of crowdfunding, they

mainly focused on one modality, and only a few studies have analyzed the multimodality of crowdfunding projects with limited influential features. While one stream of the literature has paid attention to explanatory study, it either considers only the statistics of texts, images, and videos or manually labels the visual content of a small sample. For example, Carradini and Fleischmann (2023) investigated the effects of the numbers of images, links, gifs, videos, and galleries on crowdfunding success. They found that images, links, and video presence have a positive impact, whereas gifs and galleries do not. Chen et al. (2024a) explored how linguistic styles of speech text and acoustic features of speech voice impact crowdfunding success. Their research indicates that emotional cues, both vocal and textual, significantly enhance fundraising success. Copeland (2015) observed that the textual rhetoric is less effective, shows less enthusiasm, and is less important to the campaigns than the motive and the human message of videos, when fundraising a new church on Indiegogo. Grebelsky-Lichtman et al. (2018) analyzed verbal and nonverbal immediacy and nonimmediacy communication behavior in crowdfunding videos. They found that verbal immediacy communication through videos has a positive effect on fundraising success, whereas verbal and nonverbal nonimmediacy has a negative effect. Hou et al. (2019) extracted emotional features from descriptive texts and emotions and objects from title images, but they only discussed the effects of title images on a crowdfunding platform. Lee and Park (2020) applied deep learning and topic modeling to derive topical and emotional features from unstructured visual and textual data of donation-based crowdfunding. They found that observing a small recipient in the images leads to a strong willingness to assist, positive emotions may reduce potential donors' empathy, and complex project descriptions require greater effort to understand recipients' situation and hence may deter potential donors. Yang et al. (2020) also conducted an explanatory study about the effects of text length and the number of images on pledged amount and the number of backers. Zihagh et al. (2024) examined how textual

and visual brand elements influence the fundraising success of aftermarket campaigns, concluding that prominent branding positively correlates with campaign outcomes. Additionally, Zhao et al. (2022a) examined how emotions are expressed in text and image modalities on a charity crowdfunding platform. Their experiment results show the superiority of the verbal modality in explaining crowdfunding campaign success when emotions are expressed via both text and image modalities.

Another stream of research focuses on predictive analysis, where researchers explore whether features from multimodality can improve predictive utility. It mainly relied on deep learning models to extract linguistic and visual content features from multimodal data. Al-Qershi et al. (2022) extracted linguistic features from texts and content (objects and humans) and emotions from videos, discovering that humans contribute more than objects in predicting fundraising success, and words related to experience or perception are more important than cognitive words. Blanchard et al. (2023) emphasized that visual count and image details are two top predictors of crowdfunding success, and image detail features, such as color, matter a lot. Chen et al. (2024b) used deep learning techniques to analyze the text, images, and video promotional content of travel crowdfunding projects. Their findings indicate that key predictors of success include positive emotions, pro-social engagement, cognitive vocabulary, an increase in video scenes, and rapid visual variation. Cheng et al. (2019) applied bag-of-words and word embedding (GloVe) to represent textual features and used a pre-trained VGG-16 model to extract visual features from crowdfunding images. Their experiment results show that the image features could improve success prediction performance, particularly for project profiles with little text information. Kaminshi and Hopp (2019) built a predictive model using transformed text, speech, and video content, finding that a model with text, speech, and video features achieved the best performance in predicting campaign success or failure. Perez et al. (2020) extracted

sentiment, word importance, and named entities from descriptive texts and proposed a pre-trained ResNet-152 model to extract emotion (eight types, e.g., sadness, fear, and amusement), appearance (a color or an object), and semantic features (the logit presence of predetermined objects in each image) from images to predict fraud in crowdfunding projects. They found that textual data alone provide better performance in predicting fraudulent projects. Shi et al. (2021) used VGGish deep learning architecture to extract audio features and incorporated them into the analysis of crowdfunding success. Tang et al. (2023) proposed a deep cross-attention network to extract information from videos and texts. Their experiments show improvement in performance over baseline models. Furthermore, they explored various fusion strategies involving different modalities. Zhang et al. (2021) applied random forest as a fusion classifier to validate the effectiveness of textual and visual features from texts and cover images. Zhang and Lau (2024) examined the predictive effectiveness of descriptive text, audio clips, and video clips. Their experiment results show that textual features achieve relatively high prediction accuracy when a single modality of features was used, and the combination of three modalities results in the best predictive accuracy.

2.4 Research Gaps

Our literature review reveals some gaps in the research regarding crowdfunding success prediction and the use of multimodal data geared to the crowdfunding context. First, the existing literature has predominantly investigated limited features from multimodal data, concentrating on the emotional elements of verbal and visual modalities while overlooking the ideational and textual metafunctions of multiple modalities. Due to the high dimensionality of multimodality, limited features may lead to biased and arbitrary conclusions. We posit that this gap can be mitigated by a fundamental theory that supports building a sound prediction framework for leveraging multimodal features appropriately. On the grounds of the SFL theory, we conceptualize a rational theoretical foundation

for analyzing multimodality of crowdfunding. Second, multiple data modalities, such as text, image, video, and audio, cooperate complementarily and synergistically to shape a cohesive and meaningful context. Solely capturing the contents of multimodality using deep learning models may lose vast valuable information regarding the organization and collaboration among multiple modalities. For example, layouts, sizes, movements, volumes, and contrasts of texts, images, and videos are crucial to capture viewers' attention. It is therefore necessary to model the intersemiotic complementarity³ (Royce 2007) among the multiple modalities to synergize multimodal data for crowdfunding success prediction. To bridge these gaps, we propose a framework that delves into features representing the ideational, interpersonal, and textual metafunctions of multimodal data in reward-based crowdfunding. Table 1 contrasts our study and existing studies related to multimodal data analysis on crowdfunding, highlighting the distinctive approach and contributions of our work.

Table 1. Studies related to multimodal data analysis on crowdfunding

Study	Target variable	Analysis type	Features	Modalities	Gap
Al-Qershi et al. 2022	Success or not	Predictive	Linguistic features from texts; visual contents (number of humans, facial emotions) of videos	Text and video	Only investigated the objects and emotions from texts and videos, ignoring the experiential meaning, logical meaning, and compositional meaning of multimodality (ideational and textual metafunctions)
Blanchard et al. 2023	Success or not	Predictive	Number of images, number of videos, image color, composition, figure-ground relationships, visual scene characteristics from images; analytical language, clout/persuasiveness, authenticity, and emotional tone from texts	Text and image	Overlooked the effects of experiential meaning and logical meaning of texts and images, as well as social relationship between entrepreneurs and backers (ideational and interpersonal metafunctions)
Carradini and	Success or not	Explanatory	The numbers of images, links, gifs, videos,	Image and video	Only examined the numbers of images and videos, overlooking the

³ Intersemiotic complementarity was proposed by semiotic theorist Terry Royce. It involves the idea that different forms of signs or expressions, such as words, images, sounds, or gestures, can complement each other in conveying meaning.

Fleischmann 2022			and galleries		linguistic and visual features of multimodality
Chen et al. 2024	Success or not	Explanatory	The emotional cues of fundraiser's speech; emotional cues of texts	Text and audio	Only examined acoustic emotional cues and textual emotional cues of audio and texts, overlooking ideational and textual metafunctions of multimodality
Chen et al. 2024	Success or not	Predictive	Emotions, pro-social engagement, cognitive vocabulary, increased video scenes, and rapid visual variation	Text, image, and video	Only examined emotions, pro-social engagement of texts and videos (interpersonal metafunction), and tone and content features of images (textual metafunction), overlooking the ideational metafunctions of multimodality
Cheng et al. 2019	Success or not	Predictive	Bag-of-words representation of texts; word embedding representation of texts; CNN representation of images	Text and image	Overlooked the effects of social relations and interactions conveyed by texts and images (interpersonal metafunction) and the compositional features of texts and images (textual metafunction)
Copeland 2015	Raised funds	Explanatory	Textual rhetoric; quality videos	Text and video	Overlooked the effects of experiential meaning and compositional meaning (ideational and textual metafunctions) of texts and videos
Grebelsky-Lichtman et al. 2018	Success or not	Explanatory	Verbal and nonverbal immediacy features	Video	Small sample size (120) with human coding; overlooked the experiential meaning, logical meaning, and compositional meaning of multimodality (ideational and textual metafunctions)
Hou et al. 2019	Images' emotion; the number of backers; fundraising amount	Explanatory	Texts and title images' emotion; title images' composition; title images' color; body-ground relationship; the presence of text, humans, and animals in images	Text and image	Only investigated features of title images, overlooking the effects of other images; extracted only emotional features from texts, overlooking the experiential meaning, logical meaning, and compositional meaning of multimodality (ideational and textual metafunctions)
Kaminshi and Hopp 2019	Success or not	Predictive	Word vectors of texts and speech; objects from videos	Text and video	Overlooked the effects of social relations and emotions (interpersonal metafunction) and compositional features of multimodality (textual metafunction)
Lee and Park 2020	Pledged amounts	Explanatory	Text topic distribution; humans and their emotion and gender in images	Text and image	Overlooked the effects of experiential meaning and compositional meaning (ideational and textual metafunctions) of images and interpersonal and textual metafunctions of texts
Perez et al. 2020	Fraud or not	Predictive	Emotions of texts; named entity and word importance of texts; color, texture, and	Text and image	Overlooked the effects of experiential meaning and logical meaning of texts and images (ideational metafunction)

			objects of images; face appearance in images		
Shi et al. 2021	Success or not	Predictive	Number of words, number of sentences, and sentiment of texts; audio features	Text and audio	Ignored the effects of images and overlooked the effects of experiential meaning and compositional meaning (ideational and textual metafunctions) of texts and audio
Tang et al. 2023	Success or not	Predictive	Visual features of video frames; word embedding from texts	Text and video	Ignored the effects of images and overlooked the effects of experiential meaning and compositional meaning (ideational and textual metafunctions) of texts and video
Yang et al. 2020	Fundraising amount; the number of backers	Explanatory	Text length; number of images	Text and image	Only investigated the effects of text length and the number of images on fundraising amount and the number of backers, ignoring three metafunctions of multimodality
Zihagh et al. 2024	Success or not of aftermarket campaigns	Explanatory	Textual and visual brand prominence, cognitive language style, emotional language style	Text and image	Only investigated the presence of brand in text or brand name/logo in the thumbnail and text narratives, ignoring three metafunctions from images and ideational metafunction of texts
Zhang et al. 2021	Success or not	Predictive	92 LIWC features from texts; aesthetic and technical scores for each cover image; number of faces, beauty, smiles, and emotions of faces in the cover image	Text and image	Only investigated facial attributes in the cover image and word categories from texts, overlooking ideational and textual metafunctions of texts and images
Zhang and Lau 2024	Fundraising amount; Success or not	Explanatory and predictive	Word embeddings; textual signal of video and audio instead of RGB image and sound frequency	Text, video, and audio	Only investigated semantic information of video frames and audio, overlooking the textual metafunctions (such as color, brightness, pitch, and frequency)
Zhao et al. 2022	Success or not	Explanatory	Sentiments and emotions of texts and images	Text and image	Only examined the emotions of image and text modalities, overlooking the experiential meaning, logical meaning, and compositional meaning of texts and images (ideational and textual metafunctions)
This study	Success or not	Predictive	Word embeddings; image embeddings; video embeddings; MFCC; social relation, polarity modality, lexical density, grammatical cohesion, and clarity of texts; facial expression, objects, social distance, color, and composition	Text, image, video, and audio	Comprehensively captured ideational, interpersonal, and textual metafunctions of each modality and combinations of multiple modalities, based on sound theoretical foundation and design rationales

			of images and videos; volume variation and music, speech, salience appearance time and variation of audio		
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3 Theoretical Foundation and Design Rationales

To effectively leverage multimodal data, that is, linguistic, visual, and acoustic features of texts, images, and videos, for reward-based crowdfunding success prediction, we build a theoretical foundation for the rational design of the multimodal prediction framework. While multimodality theories have rarely been observed in the IS area, they have seen extensive use in the areas of communication and media studies. Origins in systemic functional linguistics (SFL; Halliday 1985), social semiotic multimodality (SSM) approaches (Kress and Van Leeuwen 1996; O'Halloran 2008a), and multimodal discourse analysis (Scollon and Wong Scollon 2004) have gained prominence in social semiotics and discourse analysis. SSM has usually been used to analyze how meaning is created and communicated through diverse semiotic resources, such as gesture, image, music, and layout, extending beyond mere verbal language. Considering the reward-based crowdfunding context, which is a blend of various semiotics, we posit that the SFL and SSM frameworks are particularly apt for developing our predictive model. These approaches allow a comprehensive understanding of how various semiotic elements collectively contribute to the success of crowdfunding initiatives.

The SFL theory proposes that human language embodies three metafunctions, namely, the *ideational* metafunction (“constructing a model of experience”), the *interpersonal* metafunction (“enacting social relationship”), and the *textual* metafunction (“creating relevance to context”), which contribute to the semantic organization of natural languages (Halliday 1985).

The ideational metafunction focuses on the content of discourse, capturing the sequences of processes (processes of doing and happening, sensing and saying, and being and having), the participants involved in these processes, and the associated circumstances (time, space, manner, or cause of

processes; Halliday 1985). In the context of reward-based crowdfunding, the ideational metafunction discloses the fundamental components involved. Processes, participants, actions, events, entities, and circumstances collectively represent the essential aspects of the external reality in each project. The availability of rich information concerning a product's or service's design or usage processes, participants (users), actions (functions), and circumstances (relevant scenes) enhances the ability to predict fundraising outcomes. The ideational metafunction also provides effective signals to mitigate information asymmetry between entrepreneurs and backers. Drawing from signaling theory (Spence 1973), an informed party (e.g., entrepreneurs) sends observable and costly signals to a less informed party (e.g., backers) to reveal critical information about unobservable or underlying characteristics (e.g., design, processes, participants, actions, and circumstances) to promote exchange (e.g., raising funds; Spence 1973; Courtney et al. 2017). Detailed descriptive project information (e.g., ideational metafunction) not only signifies a project's quality but also demonstrates thorough preparation and viability (Mollick 2014). Consequently, we posit that the ideational metafunction is an effective predictor of fundraising success.

The interpersonal metafunction is described as “the speaker or writer doing something to the listener or reader by means of language” and displays the social relationships and values between speakers and listeners or between writers and readers (O'Halloran 2008b; Royce 2007). For descriptions of reward-based crowdfunding projects, the interpersonal metafunction describes the social interaction between backers and entrepreneurs. It can be delivered by social connections (family, friends, females, males), sentiment, and degree of certainty presented in the description. What the entrepreneurs have done, bonding social relationships or expressing certainty signals, will lead to possible funding actions from backers. Considering the interpersonal metafunction may therefore enhance fundraising success prediction.

The textual metafunction shows the interweaving of ideational and interpersonal choices into cohesive and coherent units of meaning (Halliday 2004). The textual metafunction “enables the speaker or writer to construct texts” (Halliday 1970). In terms of the cohesion of texts, Halliday and Hasan (1976) proposed two primary types of cohesive devices: grammatical and lexical. Grammatical cohesive devices refer to the way terms are structured grammatically within a discourse, whereas lexical cohesive devices show the way terms are connected lexically within sentences. Generally, these cohesive devices play a crucial role in binding sentences together and establishing connections that extend to larger units, such as paragraphs, stanzas, or chapters (Paziraie 2013). Coherence is defined as the quality that allows a discourse to conform to a consistent representation of a world of experiences (Connor and John 1990). In written communication, it is essential for the writer to ensure that his or her ideas, intentions, and arguments are conveyed with absolute understandability and clarity to the reader (Bublitz et al. 1999). On reward-based crowdfunding platforms, entrepreneurs are advised to organize multimodal data cohesively and coherently as complete messages. They should use coherence and cohesive devices to guide the process of creating and interpreting text. For example, the subjects of projects can be thematic, given, and specific, whereas complements can be non-thematic, new, and non-specific. It is obvious that a well-organized, connected, and consistent project description is more attractive and convincing. Therefore, whether the organization of multimodal data appearing in the project webpage is coherent, cohesive, clear, and unified can be a powerful predictor of fundraising success.

Besides the application of metafunctions in linguistic communication, researchers of SFL were mainly interested in non-verbal semiotic resources, such as images and sounds, to explain what features make multimodality visually-verbally coherent. Kress and Van Leeuwen (1996) proposed that visual images fulfill the metafunctions, including the representation of the experiential world

(*representational* meaning), the interaction between participants and viewers (*interactive* meaning), and the compositional arrangement of visual resources (*compositional* meaning).

Building on this, Royce (1998) identified the concept of *intersemiotic complementarity*, which suggests that “visual and verbal modes semantically complement each other to produce a single textual phenomenon.” He introduced three elements, namely, *represented participants*, *interactive participants*, and *visual compositional features* (Royce 2007), which correspond to Halliday’s three metafunctions. Represented participants encompass all the elements or entities that are actually present in the visual content, whether animate or inanimate. Interactive participants are the participants who engage with the visual content, representing the social relations between the viewer and the visual. Compositional features denote the arrangement of visual elements and textual components, contributing to the overall structural coherence of the composition (Royce 2007; Kress and Van Leeuwen 1996).

In the context of reward-based crowdfunding, entrepreneurs are advised to use multiple images or videos to express their ideas. The ideational metafunction of images and videos enables the representation of visual participants (individuals appearing in the images or videos) and entities depicted within the visual elements of each project. The interpersonal metafunction shows the social relationships between visual participants and backers. The textual metafunction represents the elements for achieving visual cohesion in the composition of a project webpage. The integration of the three metafunctions in the verbal and visual modes allows a more comprehensive and impactful communication experience between entrepreneurs and backers.

In summary, reward-based crowdfunding basically involves the simultaneous interplay of the three metafunctions: *ideational*, *interpersonal*, and *textual*. First, potential backers seek to understand a project’s product/service usage process, the users involved, and the relevant circumstances. This

information allows them to assess the project's viability and determine whether it is worthy of their funding. Second, backers consider the emotions, attitude, and judgments embodied and evaluate the social connection they perceive with the entrepreneur of a project. This assessment helps them determine whether they can trust the entrepreneur and the ability of the entrepreneur to deliver on the promise. Finally, backers appreciate cohesive and coherent visual-verbal information. Clear and well-organized information, presented in an orderly, systematic, unified, and transparent manner, enables backers to make well-informed decisions. The three metafunctions constitute panoramic views revealing the success of reward-based crowdfunding rationally and comprehensively.

Specifically, our theoretical foundation is primarily built on three earlier frameworks specifically designed for analyzing multiple semiotic resources: Halliday's metafunctions framework of languages (1985), Kress and Van Leeuwen's functional visual design (1996), and Royce's intersemiotic complementarity of languages and visual images (1998).

4 Proposed Framework

Building on the abovementioned theoretical foundation and its reflection in the reward-based crowdfunding context, we propose a framework for reward-based crowdfunding success prediction, identifying representation features of metafunctions from the text, image, and video modalities.

4.1 Ideational Metafunction Representation

The ideational metafunction of language, as described by Halliday (1985), demonstrates how we represent experience through language. It encompasses the experiential and logical meanings between clauses, revealing the organization of content and logico-semantic relations involved (Djonov 2005; Norris and Maier 2014). Royce (1998) applied Halliday's ideational meaning to represent visual structures. He identified the represented participants, which correlate with the ideational metafunction, as all the elements or entities that are present in the visual, whether animate or inanimate (Royce

2007).

For the text modality, SFL guides us to focus on the processes involved, encompassing actions, events, or states, the associated entities, and the circumstances in which these processes occur if mentioned. Therefore, we need to extract words or sentences representing processes, participants, and circumstances from crowdfunding project description text. Such extraction can be facilitated by large language models, such as gpt-4 or gpt-3.5-turbo. Subsequently, a deep learning model, such as the transfer learning model BERT (Bidirectional Encoder Representation from Transformers; Devlin et al. 2018), can be applied to derive word embeddings.

For the ideational metafunction of images, as suggested by SFL, it should encompass all the elements or entities that are actually present in the visual. Given the efficacy of deep neural networks in extracting hierarchical features from large and unstructured datasets, deep representation features learned by neural networks can be used to capture the ideational metafunction of the image modality. Deep neural networks can extract both lower-level features, such as shapes and textures, and higher-level features, such as object parts and components. The learning capabilities of neural networks, such as convolutional neural networks (CNNs), enable the understanding and interpretation of the ideational meaning of visual contents. Krizhevsky et al. (2012) made a breakthrough in image classification using CNNs, which enhanced the descriptive capability of image representation (Liu et al. 2018). Various CNN architectures, such as AlexNet (Krizhevsky et al. 2012), VGG-VD (Simonyan and Zisserman 2014), GoogleNet (Szegedy et al. 2015), and CaffeNet (Jia et al. 2014), have been used by researchers to extract image features at different levels.

Although SFL does not explicitly define the metafunctions of the video modality, it can be conceptualized as a fusion of image streams and corresponding audio. Consequently, deep neural

networks can be used to identify the ideational metafunction of videos. For example, the 3D ResNets⁴ can be used for this purpose. 3D ResNets are 3D CNNs that are built upon the residual networks (ResNets) to encode video data into sequential embeddings, using spatio-temporal (3D) kernels (Hara et al. 2017). Leveraging the capabilities of 3D ResNets allows for extraction of spatio-temporal features from video data, facilitating a more comprehensive understanding of the video modality.

Similarly, the ideational metafunction of audio should encompass the acoustic signals. Mel-frequency cepstral coefficients (MFCCs; Poles 1977) are powerful feature representations for audio signals and are effective at capturing discriminative features of audio signals, such as the spectral features of audio. They are widely used in various audio processing tasks and automatic speech recognition applications. Therefore, leveraging MFCCs enables us to analyze and comprehend the ideational aspects embedded within audio recordings.

4.2 Interpersonal Metafunction Representation

The interpersonal metafunction is realized through the use of clauses, serving as an exchange of information or goods and services. Its primary focus lies in the enactment of social relationships between the speaker or writer and the audience or viewer within a specific communication context (Halliday 2004). At the lexicogrammatical level, interpersonal meaning includes the forms of interaction and social interplay with others, polarity (positive or negative), and modality (degree of certainty and probability; Halliday 2004; Guijarro 2010).⁵

According to SFL, the words that reveal social interaction, polarity, and modality need to be extracted from reward-based crowdfunding projects to represent the interpersonal metafunction of texts. For example, the LIWC (Linguistic Inquiry and Word Count) software (Pennebaker et al. 2015) can be

⁴ <https://github.com/kenshohara/3D-ResNets>.

⁵ The term “modality” here refers to a type of metafunction representation, different from that in “data modality”, which refers to a form of data (e.g., text, image, or video).

used to extract interaction, polarity, and modality from descriptions as interpersonal features. LIWC provides a method for analyzing emotional, cognitive, social, and honest components present in verbal and written speech samples (Tausczik and Pennebaker 2010). With its 80 language categories, LIWC has been widely used in studies examining psychological processes.

Kress and Van Leeuwen (1996) extended the metafunction theory to visual design and suggested that the interpersonal metafunction of images involves features of contact, social distance, and modality between viewers and visual participants. Contact is constructed by the gaze or facial expressions of the visual participants toward the viewers and conveys information in the form of a portrayal. Social distance is determined by perceived proximity between the visual participant and the viewer in an image, with close or long shots related to the degree of intimacy between them (Kress and Van Leeuwen 2006). Modality relates to the truth, credibility, and probability of what visual participants represent to viewers, and whether the information they present is real or unreal (Royce 2007). Pages that exhibit relatively static, ordered layouts and less variation in color tend to have higher modality, often conveying factual information (Kress and Van Leeuwen 2006).

Facial expression can be detected using some vision artificial intelligence techniques. Several vision APIs, such as Google Vision API, Microsoft Azure Computer Vision API, IBM Watson Visual Recognition, and Amazon Recognition, are available for processing and extracting facial expressions from both images and videos. Additionally, vision API can detect the facial expressions, objects, and their respective locations within each image. The social distance can be calculated by the proportion of the main object to the overall image size. Furthermore, the display and color variance among the images in each project can be reached to represent modality. Consequently, interpersonal metafunction features that contribute to the portrayal of contact, social distance, and modality within images and image streams of videos can be constructed. Since the volume variation of a video also

conveys modality, for example, viewers may feel uncomfortable if the range of volume is too wide, volume variation is also included to represent modality of videos.

4.3 Textual Metafunction Representation

The textual metafunction serves as a bridge between the ideational and interpersonal meanings. For texts, it is concerned with cohesive and coherent meaning analysis, including elements such as lexical density, grammatical cohesion, and clarity. Texts can be perceived as cohesive units operating within either the grammatical or the lexical zone of the lexicogrammatical continuum (Halliday 2004). Halliday (1996) developed a measure of lexical density as “the proportion of content (lexical) words—basically nouns, full verbs, adjectives, and adverbs derived from adjectives—over the total number of words in a text.” Additionally, he defined grammatical cohesion as conjunction, reference, and ellipsis within texts. Furthermore, clarity is the pivotal measure of the coherence of texts. It can be assessed through precise and deliberate words, clear and concise sentence structure, consistent organization, smooth transitions, and readability. Thus, lexical density, grammatical cohesion, and clarity serve as essential features to represent the textual metafunction of texts. Lexical words can be extracted from description texts, and the proportions of lexical words within the texts can be used to capture the lexical density. Large language models can be used to measure the grammatical cohesion and clarity within texts.

In visual design, Kress and Van Leeuwen (2006) proposed that achieving compositional cohesion requires attention to three factors: information value, framing, and salience of images. The information value refers to the placement of elements within an image, including their location in the center or margin, left or right. Elements on the left side of a visual image are considered as something already known, whereas the right side presents new information. The top and bottom of an image represent the realms of the ideal and the real, respectively (Kress and Van Leeuwen, 2006). To capture

this feature, the location of the main object (i.e., the largest element) in each image can be extracted using a vision API.

Framing refers to the visual devices used to connect or separate the content in an image. Devices such as borders, spacing, and color can be used to frame content (Norris and Maier 2014). Color is the key feature of image framing because it connects or separates important objects within simple pictures (Norris and Maier 2014). Saliency refers to the ability of a visual participant to capture the viewer's attention. Saliency is related to size, movement, and contrast and is determined by a variety of features, such as the number and size of elements, tonal contrast, and color contrast (Norris and Maier 2014). Therefore, we extract features of the image colors to represent framing and the number and size of main objects and the color contrast to represent the saliency of images. For videos, our analysis incorporates not only image-related features derived from image streams but also audio variations to signify the prominence of audio elements within videos. For instance, if the audio suddenly transfers from silence to speech, or from music to speech, we consider it a moment designed to attract viewers' attention. Table 2 summarizes the proposed metafunction representations of multimodality.

Table 2. Metafunction representations of multimodality

Modality		Ideational Metafunction	Interpersonal Metafunction	Textual Metafunction
Text		Processes, participants, and circumstances	Interaction and social relation; polarity; modality	Lexical density; grammatical cohesion; clarity
Image		Visual elements or entities	Contact; social distance; modality	Information value; framing; saliency
Video	Image	Visual elements or entities	Contact; social distance; modality	Information value; framing; saliency (main object appearance time included)
	Audio	MFCC representation	Modality (volume variation)	Saliency (music, speech, saliency appearance time and variation)

Figure 1 outlines the proposed framework. First, we collect multimodal data from the reward-based

crowdfunding platform and then conduct a series of data preprocessing. Next, we extract the ideational, interpersonal, and textual representation features from the multimodal data using various linguistic, visual, and acoustic processing technologies, such as deep learning models (BERT, CNNs, and 3D ResNets), large language models, MFCC, Google Vision API, and LIWC. Finally, we develop machine learning models to predict fundraising success based on the three metafunction representation feature sets extracted from the multimodal data.

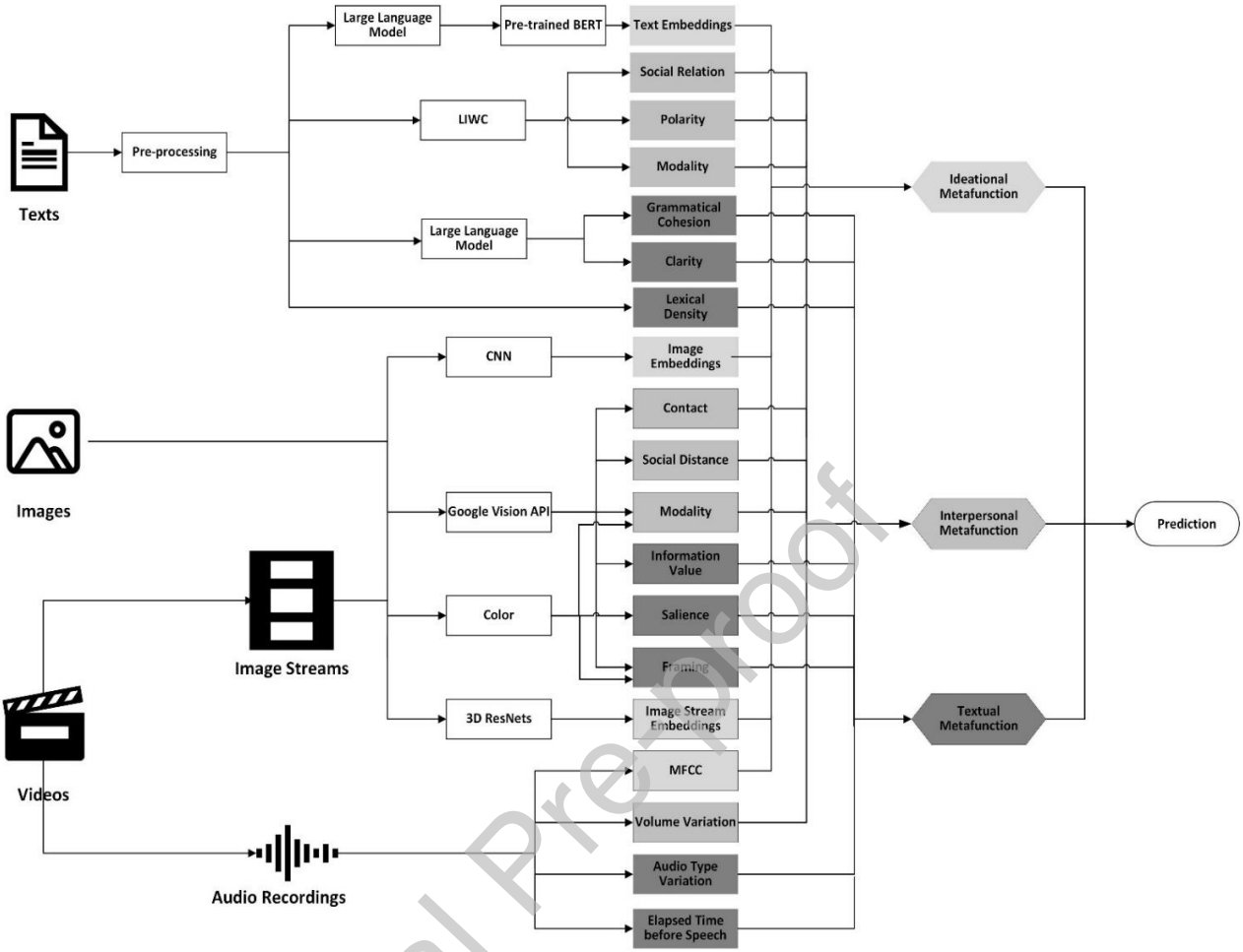


Figure 1. Outline of the proposed framework.

5 Empirical Evaluation

We have evaluated our proposed framework on a dataset collected from Kickstarter.com, a well-known reward-based crowdfunding platform.

5.1 Data

We collected data about the projects launched between 2014 and 2019 in the US market. We then kept completed (i.e., successful or failed) projects for our evaluation and eliminated uncompleted (i.e., suspended or canceled) projects. In total, there were 121,486 launched projects. Given the focus of our research on the multimodality of projects, we strove to select project categories with rich data

modalities. For example, categories such as journalism and crafts have limited video content. Thus, we opted for the music category, which ranked among the top two categories of the launched projects and offered a diverse range of modalities, for our empirical study. Among the 12,541 completed projects in the music category, we retained 8,559 projects that incorporated text, image, and video modalities for evaluation. Out of these projects, 6,164 were successful and 2,395 were not. Next, we extracted the texts, images, and videos from the home page of each project. Figure 2 illustrates the layout of texts, images, and videos on a Kickstarter project page, showcasing how these modalities are presented to viewers.

The original data contain some noise, especially in the text and video modalities. To improve the data quality, we conducted a series of preprocessing processes. For the text data, we removed stop words, numbers, and HTML tags and converted characters to lowercase. For images, we stacked all the images of each project into one image for deep representations. For videos, we selected the first video of each project for analysis, as it usually includes the main instruction of the project, which backers are most likely to watch. We then sliced each video into image sequences per second and extracted the first 180 seconds, which backers may pay the most attention to.

Besides the multimodal data, we also extracted and standardized some meta features of each project, including fundraising goal, duration, description text length, number of images, and number of videos.



Figure 2. The home page of a Kickstarter project with multimodality.

5.2 Feature Extraction

We extracted features representing the ideational, interpersonal, and textual metafunctions from the multimodal data. Descriptive statistics of the features (including the meta features) are listed in Online Appendix A.

5.2.1 Ideational Metafunction

The ideational metafunction reveals the elements or entities that are present in the multimodal data, including experiential and logical meanings. For the text modality, we extracted processes, participants, and circumstances from textual descriptions using large language models. Specifically, we defined a structured template to illustrate processes, participants, and circumstances within the text. We then used LangChain⁶ to generate prompts⁷ and directed them to the large language model (gpt-3.5-turbo-16k). Subsequently, we processed representative words or sentences denoting processes, participants, and circumstances into input embeddings (token embeddings, segment embeddings, and position embeddings) required by BERT and then initialized the BERT model with the pre-trained parameters.⁸ We used the text embedding outputs of BERT as input features of prediction models. For the image modality, we applied the VGG-16 architecture (Simonyan and Zisserman 2014),⁹ which has demonstrated superior recognition or classification accuracy in CNNs (Alom et al. 2019), to map images into deep representations. For the video modality, we extracted image stream embeddings using 3D ResNets¹⁰ and coefficients of the short-term power spectrum (MFCC) of audio recordings. Therefore, we obtained text embeddings, image embeddings, and video embeddings from deep neural network models (BERT, VGG-16, and 3D ResNets) as well as MFCC from audio recordings to represent the ideational meanings of projects. As the number of embedding features is much larger than that of interpersonal or textual representation features, we conducted feature reconstruction using long short-term memory (LSTM) attention. Some key parameter settings of the deep learning models

⁶ LangChain is an open-source framework for developing applications powered by large language models (https://python.langchain.com/docs/get_started/introduction).

⁷ Structured texts that can be interpreted and understood by a large language model.

⁸ <https://github.com/google-research/bert/blob/master/multilingual.md>.

⁹ <https://github.com/minar09/VGG16-PyTorch>.

¹⁰ <https://github.com/kenshohara/3D-ResNets>.

are listed in Online Appendix B.

5.2.2 Interpersonal Metafunction

The interpersonal metafunction reveals interaction and social relations with viewers. We obtained social relation (social, family, friends, females, males), polarity (positive or negative emotions), and modality (certainty, tentativeness, risk) of each project description text from LIWC as the interpersonal metafunction of the text.

To capture the interpersonal metafunction revealed in images, we extracted features of contact, social distance, and modality between viewers and visual participants. To represent contact, we extracted the number of faces and facial expressions (positive or negative) and calculated their statistics (i.e., maximum, minimum, mean, and standard deviation) among the images of each project. Since the size of the main human (i.e., the largest human body) compared to the whole image reflects the distance between viewers and visual participants to some extent, we extracted the coordinates of the largest human in each image and then calculated his/her square ratio to the image size to represent the long, medium, or close shot. We further calculated the relative statistics (i.e., maximum, minimum) of the ratios among the images of each project to represent social distance. Finally, since the standard deviation reflects the variation of data series, we computed the standard deviation of color (i.e., warm/cool, saturation, brightness, and contrast) among the images of each project and the standard deviation of compositions, specifically, the distance between the main human mid-point and the image horizontal/vertical central line in each project (i.e., visual balance) and the distance between the main human mid-point and the image equally-spaced horizontal/vertical lines in each project (i.e., the rule of thirds), to represent modality.

For videos, we sliced each video into image frames by each second. Similarly, we extracted representation features of contact, social distance, and modality from image frames. In addition, we

extracted some video-related features to represent the interpersonal metafunction. Specifically, we calculated the ratio of human appearance duration to video duration, the maximum and average numbers of facial expressions, and audio volume variation to represent contact.

5.2.3 Textual Metafunction

The textual metafunction reveals the organization of the structure creating coherent and cohesive texts or images. For the text modality, we first extracted lexical words (i.e., noun, verb, adjective, adverbial, and preposition) from the textual descriptions. The proportions of these lexical words within the text were used to gauge the lexical density, providing insight into the richness of the vocabulary used. We then used the large language model to evaluate the grammatical cohesion and clarity within the text. This analysis enabled us to delve deeper into the intricate grammatical relationships within the text, as well as its overall understandability and coherence.

To capture the textual metafunction of images, we extracted features of information value, framing, and salience. To represent the information value of images, we extracted the location of the main object in each image and calculated the visual balance and the rule of thirds for each image. We then used the means of these distances for each project to represent the information value. For the framing of images, we calculated the average and maximum color ratios of each image to the whole project. Since salience shows the capability of a visual participant to attract the viewer's attention, we obtained the number of objects, the square ratio of the main object to each image in each project, and the color contrast of each project to represent salience.

For the video modality, we added the elapsed time before the human appearance and the elapsed time before the speech appearance to represent information value. We extracted the total/average appearance time and frequency ratios of music/speech/salience in each video to represent salience.

5.3 Experiments

To evaluate the predictive utilities of the proposed ideational, interpersonal, and textual metafunctions of multimodal data of reward-based crowdfunding, we conducted five experiments, building classifiers using machine learning and multimodal deep learning methods. The machine learning methods are logistic regression (LR), least absolute shrinkage and selection operator (LASSO), support vector machine (SVM), random forest (RF), and XGBoost (XGB). These machine learning methods include both linear and nonlinear methods, as well as both single-classifier and ensemble methods. The multimodal deep learning methods include concatenation-based multimodal deep learning (Concat; Ngiam et al. 2011), automatic feature interaction learning via self-attentive neural networks (AutoInt; Song et al. 2019), adversarial representation learning and graph fusion network for multimodal fusion (ARGF; Mai et al. 2020), private-shared disentangled multimodal VAE (DMVAE; Lee and Pavlovic 2021), multimodal recurrent attention CNN (MRACNN; Zhang et al. 2020), the soft voting ensemble learning-based approach (SoftVoting; Salur and Aydin 2022), and margin-aware distillation and modality-aware regularization (MMANet; Wei et al. 2023).

In Experiment 1, with each machine learning classification method (i.e., LR, LASSO, SVM, RF, or XGB), we included the meta features as the baseline model and then added our proposed ideational features to the feature set to form the ideational model. We used the ideational models to assess the predictive utilities of the ideational metafunction in each modality (i.e., text, image, or video) and multimodality (i.e., combination of all three modalities). Similarly, in Experiments 2 and 3, we examined the predictive utilities of the interpersonal and textual representation features, respectively, on each modality and multimodality. In Experiment 4, we further compared the three metafunction feature sets, as well as the possible combinations of them, to examine possible interactions among the metafunctions in influencing prediction performance. In Experiment 5, we compared the multimodal

representation features extracted by our theory-guided framework (i.e., *theory-guided multimodal features*) with those extracted by suitable neural networks (i.e., *data-driven multimodal features*) across various multimodal deep learning methods (i.e., Concat, AutoInt, ARGF, DMVAE, MRACNN, SoftVoting, and MMANet) in predicting reward-based crowdfunding success.

For every setting in the experiments, we estimated the prediction performance in terms of the area under the receiver operating characteristic curve (AUC) and Kolmogorov–Smirnov (KS) statistic (Massey 1951), through ten independent rounds of tenfold cross-validations, resulting in 100 estimates for each metric.

6 Results

6.1 Experiment 1: Ideational Metafunction

Table 3 summarizes the predictive performance (in terms of AUC and KS) of the five models (i.e., baseline, text, image, video, and multimodal) learned by the five machine learning classification methods (i.e., LR, LASSO, SVM, RF, and XGB). Statistical significance test results are available in Online Appendix C. Overall, incorporating the ideational metafunction of any modality or multimodality outperformed the baseline model ($p < .001$), demonstrating the value of the ideational metafunction in crowdfunding success prediction.

Table 3. Predictive performance of the ideational metafunction

Modalities	Metric	LR	LASSO	SVM	RF	XGB
Baseline	AUC	0.690 (0.017)	0.678 (0.018)	0.669 (0.021)	0.638 (0.016)	0.710 (0.017)
	KS	0.314 (0.030)	0.303 (0.030)	0.307 (0.035)	0.235 (0.027)	0.329 (0.029)
Text	AUC	0.837 (0.015)	0.828 (0.016)	0.815 (0.015)	0.826 (0.016)	0.837 (0.014)
	KS	0.538 (0.029)	0.523 (0.029)	0.524 (0.031)	0.515 (0.030)	0.531 (0.028)
Image	AUC	0.761 (0.020)	0.734 (0.019)	0.771 (0.021)	0.761 (0.019)	0.798 (0.018)
	KS	0.411 (0.037)	0.370 (0.036)	0.433 (0.036)	0.405 (0.034)	0.461 (0.035)
Video	AUC	0.816 (0.016)	0.794 (0.017)	0.795 (0.018)	0.813 (0.016)	0.828 (0.016)
	KS	0.503 (0.032)	0.472 (0.031)	0.494 (0.031)	0.495 (0.031)	0.517 (0.031)
Multimodal	AUC	0.879 (0.014)	0.863 (0.014)	0.862 (0.014)	0.870 (0.014)	0.882 (0.012)
	KS	0.599 (0.029)	0.579 (0.028)	0.588 (0.028)	0.579 (0.029)	0.606 (0.026)

Note: Standard deviations are enclosed in parentheses.

When the predictive effectiveness of different modalities was compared, combining the ideational

metafunctions of the multimodal data outperformed any single data modality ($p < .001$). This implies that synthesizing data modalities helps to improve predictive performance. Furthermore, when the predictive utility of a single modality was examined, the ideational metafunction of the text modality outperformed those of the image and video modalities ($p < .001$). This finding indicates that text is a more effective modality for conveying the ideational metafunction to predict fundraising success. Since the ideational metafunction represents entrepreneurs' logics and experiences, our results suggest that text is a more effective medium for the transmission of logical constructs and experiential narratives in reward-based crowdfunding.

6.2 Experiment 2: Interpersonal Metafunction

As shown in Table 4 (results for the baseline model are the same as those in Experiment 1 and hence omitted), incorporating the interpersonal metafunction of any data modality or multimodality outperformed the baseline model ($p < .001$). This highlights the value of social relationships in fundraising success prediction. When the interpersonal metafunctions of different data modalities were compared, multimodality again outperformed any single modality ($p < .001$), indicating the superior predictive effectiveness of multimodality on the interpersonal metafunction.

Interestingly, we observed no statistically significant difference between the predictive effectiveness of the image model and that of the video model, while either the image model or the video model outperformed the text model ($p < .001$). This implies that the interpersonal metafunction conveyed by the video modality contributed almost as much as that conveyed by the image modality to the predictive effectiveness and was more valuable than the text modality. This indicates that for fostering social relationships and connections, communication through images or videos is more effective than that through text alone.

The interpersonal metafunction refers to the resource for “meaning as a form of action” (Halliday

1985), where the speaker or writer does something to the listener or reader to build a social relationship through communication. Prior research has identified that pictures are associated with proximity, while words with distance. As Torrez et al. (2019) suggested, communicators may use visual messages as a way to decrease social distance and a method for invoking proximity. People prefer visual over verbal means of communication when aiming to signal a proximity orientation toward the recipient (Amit et al. 2013). This could explain why videos and images perform better than text in conveying interpersonal signals, as they naturally foster a sense of immediacy and social connection.

Table 4. Predictive performance of the interpersonal metafunction

Modalities	Metric	LR	LASSO	SVM	RF	XGB
Text	AUC	0.744 (0.020)	0.723 (0.020)	0.755 (0.020)	0.762 (0.020)	0.773 (0.018)
	KS	0.397 (0.033)	0.363 (0.033)	0.404 (0.034)	0.403 (0.034)	0.427 (0.033)
Image	AUC	0.778 (0.019)	0.752 (0.018)	0.773 (0.018)	0.782 (0.017)	0.795 (0.019)
	KS	0.442 (0.034)	0.400 (0.031)	0.436 (0.031)	0.442 (0.028)	0.466 (0.034)
Video	AUC	0.777 (0.019)	0.752 (0.018)	0.772 (0.018)	0.783 (0.018)	0.795 (0.019)
	KS	0.442 (0.034)	0.400 (0.031)	0.435 (0.031)	0.444 (0.031)	0.465 (0.034)
Multimodal	AUC	0.792 (0.018)	0.772 (0.017)	0.796 (0.018)	0.797 (0.018)	0.811 (0.017)
	KS	0.463 (0.033)	0.431 (0.031)	0.472 (0.033)	0.466 (0.035)	0.488 (0.032)

Note: Standard deviations are enclosed in parentheses.

6.3 Experiment 3: Textual Metafunction

The results of Experiment 3 (summarized in Table 5) show that incorporating the textual metafunction of any data modality or multimodality outperformed the baseline model ($p < .001$). Multimodality again outperformed any single modality ($p < .001$), showing that increasing the data modalities contributes to better prediction performance. Comparing the effectiveness of single modalities, the textual metafunction of videos overperformed those of texts ($p < .001$) and images ($p < .001$), whereas there was no significant difference between texts and images. This implies that the textual metafunction conveyed by videos was more valuable in predicting fundraising success than those conveyed by texts or images. Given that the textual metafunction represents the structural cohesion and compositional arrangement of verbal and visual elements, video as a medium uniquely enhances

these semiotic aspects by seamlessly integrating them. This allows videos to convey verbal cohesion and visual composition more effectively than static images or text alone. The dynamic nature of videos enriches the interplay between these modalities, providing a more cohesive and contextually enriched medium for communication. This could explain why videos outperform text or images in conveying the textual metafunction.

Table 5. Predictive performance of the textual metafunction

Model	Metric	LR	LASSO	SVM	RF	XGB
Texts	AUC	0.727 (0.020)	0.701 (0.020)	0.740 (0.021)	0.760 (0.020)	0.774 (0.020)
	KS	0.362 (0.034)	0.320 (0.035)	0.392 (0.037)	0.407 (0.034)	0.433 (0.034)
Image	AUC	0.736 (0.020)	0.708 (0.019)	0.729 (0.020)	0.742 (0.021)	0.762 (0.019)
	KS	0.370 (0.035)	0.338 (0.034)	0.365 (0.034)	0.370 (0.035)	0.410 (0.034)
Video	AUC	0.776 (0.019)	0.750 (0.018)	0.774 (0.018)	0.788 (0.019)	0.798 (0.018)
	KS	0.439 (0.033)	0.404 (0.031)	0.434 (0.031)	0.455 (0.031)	0.473 (0.035)
Multimodal	AUC	0.787 (0.018)	0.764 (0.018)	0.785 (0.019)	0.800 (0.017)	0.816 (0.018)
	KS	0.455 (0.033)	0.414 (0.032)	0.462 (0.036)	0.470 (0.031)	0.503 (0.035)

Note: Standard deviations are enclosed in parentheses.

6.4 Experiment 4: Combinations of Metafunctions

Table 6 summarizes the predictive performance of combinations of metafunctions on the multimodal data (results for the three metafunctions individually are the same as those in Experiments 1 to 3 and hence are omitted). The results show that combining any two metafunctions outperformed each of the two metafunctions alone ($p < .001$), highlighting the importance of considering multiple metafunctions in predicting fundraising success. Furthermore, the combination of all three metafunctions outperformed any combination of two metafunctions ($p < .001$). This implies that all three metafunctions of multimodal data in crowdfunding interact complementarily to generate superior prediction performance.

Table 6. Predictive performance of metafunction combinations

Model	Metric	LR	LASSO	SVM	RF	XGB
Idt+Ips	AUC	0.886 (0.014)	0.875 (0.014)	0.875 (0.014)	0.877 (0.014)	0.895 (0.012)
	KS	0.619 (0.027)	0.601 (0.027)	0.606 (0.028)	0.596 (0.027)	0.627 (0.024)
Idt+Txt	AUC	0.888 (0.013)	0.877 (0.014)	0.873 (0.014)	0.876 (0.014)	0.895 (0.012)
	KS	0.619 (0.027)	0.599 (0.027)	0.603 (0.029)	0.595 (0.028)	0.627 (0.027)
Ips+Txt	AUC	0.812 (0.018)	0.791 (0.017)	0.809 (0.018)	0.815 (0.017)	0.837 (0.016)
	KS	0.490 (0.032)	0.452 (0.030)	0.491 (0.033)	0.488 (0.030)	0.520 (0.033)
All	AUC	0.890 (0.014)	0.878 (0.014)	0.878 (0.014)	0.877 (0.014)	0.907 (0.012)
	KS	0.625 (0.028)	0.604 (0.027)	0.606 (0.029)	0.599 (0.027)	0.633 (0.025)

Note: Standard deviations are enclosed in parentheses. Idt: Ideational; Ips: Interpersonal; Txt: Textual.

6.5 Experiment 5: Theory-Guided versus Data-Driven Multimodal Features across

Multimodal Deep Learning Methods

To further assess the effectiveness of the theory-guided multimodal features extracted by our framework in predicting reward-based crowdfunding success, we compared our proposed theory-guided multimodal model with a standard data-driven model, using various multimodal deep learning methods (i.e., Concat, AutoInt, ARGF, DMVAE, MRACNN, SoftVoting, and MMANet). In the data-driven model, we used embeddings for text, image, and video content. The results (summarized in Table 7) show that our theory-guided model, which integrates metafunctional representative features derived from theoretical constructs, consistently outperformed the purely data-driven multimodal features ($p < .001$) across various multimodal deep learning methods. This highlights the effectiveness of the theoretical framework in enhancing multimodal prediction performance, demonstrating the necessity and efficacy of adopting metafunctions to better leverage multimodal data in the reward-based crowdfunding context.

Table 7. Predictive performance of theory-guided versus data-driven multimodal features

Method	Data-driven		Theory-guided	
	AUC	KS	AUC	KS
Concat	0.752 (0.016)	0.504 (0.033)	0.826 (0.016)	0.653 (0.032)
MARCNN	0.751 (0.018)	0.501 (0.036)	0.825 (0.015)	0.650 (0.031)
ARGF	0.722 (0.019)	0.444 (0.039)	0.809 (0.021)	0.617 (0.041)
AutoInt	0.745 (0.018)	0.490 (0.035)	0.820 (0.017)	0.640 (0.035)
DMVAE	0.749 (0.017)	0.499 (0.041)	0.815 (0.065)	0.630 (0.013)
MMANet	0.758 (0.026)	0.516 (0.051)	0.834 (0.010)	0.669 (0.020)
SoftVoting	0.747 (0.046)	0.494 (0.092)	0.829 (0.021)	0.658 (0.039)

Note: Standard deviations are enclosed in parentheses.

7 Discussion

Our experiments yielded some intriguing findings. First, identifying ideational, interpersonal, and textual metafunctions of multimodal data indeed helped to improve the performance of fundraising success prediction. This finding reveals that experiential and logical meaning, social interaction, and the composition of the project description page have predictive values. Second, a comparison of the predictive utilities of the three metafunctions revealed that the ideational metafunction outperformed the interpersonal metafunction and the textual metafunction. This highlights the important role of processes, participants, and circumstances of a project in predicting fundraising success. Moreover, the combination of all three metafunctions outperformed any single metafunction or other combination of metafunctions. This verifies our suggestion that the three metafunctions interplay complementarily and synergistically in predicting fundraising success.

In addition, our results show the superior predictive utility of multimodal data over any single modality on any metafunction, implying that increasing the diversity of data modalities is valuable for predicting fundraising success. With regard to the predictive utilities of various modalities, the interpersonal and textual metafunctions conveyed by videos were more valuable than those conveyed by texts in predicting fundraising success. This implies that social connections or interactions and compositions are more effectively delivered through more active and engaged modalities, such as videos. Visual elements are more emotionally communicative than textual elements. In contrast, the

text modality demonstrated superior predictive value in conveying the ideational metafunction. This reflects that texts are better than images or videos at delivering logical and structured meanings, such as processes, participants, or circumstances, when predicting reward-based crowdfunding success. Interestingly, we found that the image modality showed almost the same predictive utility as the video modality in conveying the interpersonal metafunction. This implies that the image modality has good enough predictive power in expressing social interactions and relations, and the additional movements in videos may not further improve the prediction performance significantly.

8 Contributions and Implications

Our study has implications for both research and practice. For research, it makes a pioneering effort at proposing a theory-guided framework for multimodality-based prediction. Specifically, we adopted the metafunctions framework of languages (Halliday 1985), functional visual design (Kress and Van Leeuwen 1996), and intersemiotic complementarity of languages and visual images (Royce 1998) to discover metafunctions of multimodal data of reward-based crowdfunding projects. This theoretical approach has enabled us to uncover the underlying metafunctions present within the multimodal data of projects. Through empirical evaluation, we have demonstrated the superior effectiveness of the ideational, interpersonal, and textual metafunctions over purely data-driven multimodal features in predicting fundraising success. Moreover, we have demonstrated the predictive utility of any of the metafunctions and metafunction combinations. Second, our study is among the first to pay attention to the complementarity among multiple modalities to build a cohesive meaning. Besides the content conveyed by multiple modalities, the cohesion and coherence of multimodality have been found to contain valuable predictive information. Third, our research has revealed that adding modalities of data significantly enhances predictive performance. Furthermore, we have discovered intriguing patterns concerning the metafunctions conveyed by different data modalities in reward-based

crowdfunding success prediction. Specifically, our empirical results have highlighted a strong impact of the interpersonal and textual metafunctions on the video modality, contrasted with a comparatively weak effect on the text modality. In contrast, we have discovered a superior effect of the ideational metafunction conveyed by the text modality over that conveyed by the video or image modality. These insights contribute to advancing our understanding of the complementary interaction among multimodal elements and their roles in predicting reward-based crowdfunding success.

For practice, our findings suggest the importance for entrepreneurs of diversifying their data by incorporating multiple modalities. Moreover, since different modalities are more effective at conveying different metafunctions, entrepreneurs should thoughtfully select the appropriate modalities to enhance predictive utilities of specific metafunctions. For example, when entrepreneurs aim to articulate logical and experiential ideas, the text modality may provide more valuable information to predict success. On the other hand, if a project is immersive or perceptual in nature, the video modality may be more effective in improving success prediction performance.

Our proposed framework is based on the home pages of projects. It aims to help stakeholders effectively predict campaigns' success once they are launched. Entrepreneurs need to understand the importance of earlier pledges and momentum throughout the campaign process (Shneor et al. 2020). Our predicted outcomes can assist them in understanding the prospect of their campaigns and in adjusting their campaign strategy at the beginning or during the fundraising period. For backers, our proposed framework provides more information about the potential success of a campaign and helps them reduce the risk of investing in campaigns that are unlikely to meet fundraising goals. Effective predictions help backers make informed investment decisions in advance, reducing their opportunity costs and maintaining their funding returns. For reward-based crowdfunding platforms, effective predictions can help to build up a sustainable platform (Shneor et al. 2020) and keep users engaged

with a platform, since the platform can provide them with valuable advice about campaign potentials, helping them make informed decisions, and hence attracting more users and increasing the overall participation on the platform. Additionally, platforms can provide a more user-friendly website experience by recommending promising campaigns, which can help to develop a strong reputation for intelligence and reliability.

9 Conclusion

With the prevalent emergence of multimodal data on websites, entrepreneurs are increasingly leveraging a variety of modalities to sell their ideas and make a good impression. However, a systematic framework for studying how multimodal data collaboratively complement and synergize with each other, to conduct a predictive task, is still lacking. Our study opens up new avenues for analyzing the predictive effectiveness of multimodality by proposing a robust theory-guided framework. We have identified the metafunctions of multimodality in reward-based crowdfunding and demonstrated their significant predictive value.

Despite the insights provided by our study, it has some limitations, which may be addressed in future research. First, although we incorporated some quality-related features, such as lexical density, clarity, and grammatical cohesion, we did not directly assess the actual quality of textual, image, or audio elements. Similarly, our analysis did not account for the conceptual coherence between modalities (e.g., how the sound effects are relevant to the images in a video or the text presented) or the quality of the arguments presented via various modalities (e.g., how convincing, interesting, well documented, and relevant the arguments made are). All these issues are likely to influence the likelihood of success, and their effects may interact with the aspects that are coded and analyzed. Future research may further extend our work in these areas. Second, our empirical evaluation primarily focused on projects in the US market within the music category on Kickstarter. Campaigns from different countries may

have inherent cultural differences in various aspects, such as visual aesthetics and communication styles. Campaigns related to music projects may be particularly prone to the use of audio elements versus others and may have special characteristics that may not characterize other types of projects (e.g., technology). Moreover, while data from Kickstarter have been used extensively in many other studies for running success prediction analyses, it is an outlier platform by size and scope, and hence may not generalize to reflect the dynamics of the majority of much smaller platforms, which mostly cater to local markets. To enhance the robustness of our findings, future research may assess the utility of metafunctions across various project categories, on more reward-based crowdfunding platforms, and in diverse cultural settings. Third, while we focused on reward-based crowdfunding in this paper, our proposed framework may be adapted to other crowdfunding models and even other e-commerce websites that have multimodal data posted on one page. Future research may examine the effectiveness of our framework in these other contexts.

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